Module 3: Binomial and Poisson Models

Lesson 4 | Select a count model and calculate the right event

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What we will learn

A named probability distribution connects a set of assumptions to a probability mass function. We use a **binomial model** for successes in a fixed number of trials and a **Poisson model** for occurrences within a fixed exposure interval.

By the end, you should be able to justify the model, translate a verbal event into an inequality, calculate its probability in R, and interpret simulation as a model illustration.

Reading: main textbook, §5.5, printed pp.242–249 (Martin Clothing Store, p.244), and §5.6, pp.252–255 (arrivals example, p.253). The hypergeometric distribution in §5.7 is a later extension for sampling without replacement; do not assume a binomial model automatically applies there.

```
# library(): loads a package so its functions can be used in this session.
# theme_set(): sets the default theme for subsequent ggplot charts in this R session.
# theme_minimal(): uses a minimal chart style; base_size controls the base text size in
 ↪  points.
# Load display tools and choose a common chart theme; calculations use base R
 ↪  distributions.
# No observation file is imported: the following examples use stated textbook model
 ↪  parameters.
library(ggplot2)
library(knitr)
theme_set(theme_minimal(base_size = 13))
```

Business case: Martin Clothing Store purchases

At **Martin Clothing Store**, the book asks about the next **three customers**. The stated probability that a customer purchases is **0.30**. Assume that probability is the same for each customer and that the three purchase decisions are independent. What is the probability that exactly two customers purchase? Each trial concerns one customer; the random variable counts purchases among the three customers.

في متجر **Martin Clothing Store**، يتناول سؤال الكتاب **الزبائن الثلاثة القادمين**. احتمال أن يشتري الزبون هو **0.30** وفق المعطيات. نفترض ثبات هذا الاحتمال لكل زبون واستقلال قرارات الشراء الثلاثة. ما احتمال أن يشتري زبونان بالضبط؟ تتعلق كل محاولة بزبون واحد، ويعدّ المتغير العشوائي المشترين بين الزبائن الثلاثة.

Check the binomial assumptions

Let X count purchases. A binomial model requires:

1. A fixed number of trials, $n = 3$.
2. Two outcomes on each trial: purchase or no purchase.
3. The same success probability on every trial, $p = 0.30$.
4. Independent trial outcomes.

“Success” is a mathematical label for the outcome being counted. It need not mean a desirable outcome in other applications. A shopping group making a joint purchase decision, or a promotion that changes during observation, could violate these assumptions.

We write $X \sim \text{Binomial}(n, p)$, with

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \dots, n.$$

Exactly two purchases can occur in three arrangements: purchase–purchase–no purchase, purchase–no purchase–purchase, or no purchase–purchase–purchase. Each arrangement has probability $0.3^2(0.7)$, so the total is $3(0.3^2)(0.7) = 0.189$.

```
# data.frame(): builds a table whose named columns have compatible lengths.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# dbinom(x, size, prob) returns P(X=x); choose(3,2) counts arrangements of two successes.
# size is the fixed number of customers, while prob is the probability for one customer.
binomial <- data.frame(Purchases = 0:3, Probability = dbinom(0:3, size = 3, prob = 0.30))
kable(binomial, digits = 3)
```

Purchases	Probability
0	0.343
1	0.441
2	0.189
3	0.027

```
stopifnot(abs(dbinom(2, 3, .30) - choose(3, 2) * .30^2 * .70) < 1e-12)
```

Exact, cumulative, and upper-tail probabilities

```
# c(): combines values into a vector, converting them to a common type when needed.
# dbinom(): returns binomial point probabilities for size trials and success probability
 ↪  prob.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
```

```r
# data.frame(): builds a table whose named columns have compatible lengths.
# names(): reads or assigns names to vector elements or table columns.
# unname(): removes names or dimension names without changing the underlying values.
# pbinom(k,...) gives P(X<=k); lower.tail=FALSE instead gives P(X>k).
# At least two means X>1, so its upper-tail call uses 1 rather than 2.
answers <- c("Exactly two" = dbinom(2, size = 3, prob = .30),
             "At most two" = pbinom(2, size = 3, prob = .30),
             "At least two" = pbinom(1, size = 3, prob = .30, lower.tail = FALSE),
             "At least one" = pbinom(0, size = 3, prob = .30, lower.tail = FALSE))
kable(data.frame(Event = names(answers), Probability = unname(answers)), digits = 3)
```

Event	Probability
Exactly two	0.189
At most two	0.973
At least two	0.216
At least one	0.657

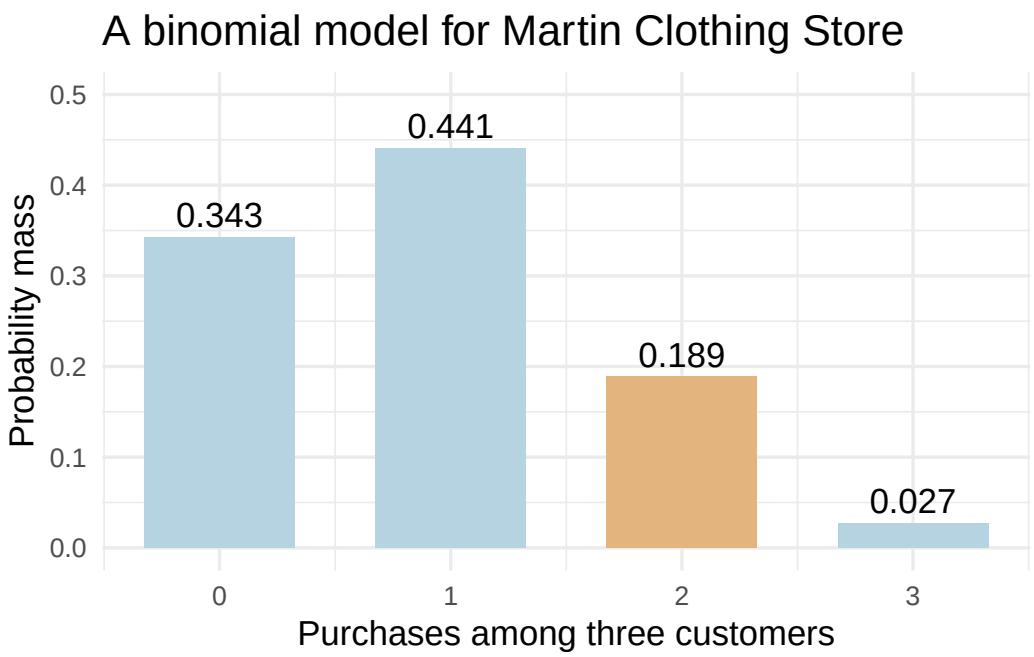
The probability of exactly two purchases is **0.189**; at least two has probability **0.216**. Using `pbinom(2, ..., lower.tail=FALSE)` would calculate more than two, which is a different event.

For a binomial count,

$$E(X) = np = 0.9, \qquad \text{Var}(X) = np(1-p) = 0.63.$$

The SD is $\sqrt{0.63} \approx 0.794$ purchases. The expected count 0.9 is an average across repetitions, not the purchase probability for a single customer and not a possible count for one three-customer group.

```r
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_col(): draws bars at supplied heights; it does not count records.
# geom_text(): draws labels from mapped data; position and hjust/vjust control placement.
# sprintf(): formats text using placeholders such as %.2f for two decimal places.
# scale_fill_manual(): assigns explicit fill colors to categories; named values preserve
 ↪  category-color mappings.
# c(): combines values into a vector, converting them to a common type when needed.
# scale_x_continuous(): controls numeric x-axis breaks, labels, limits, and spacing.
# scale_y_continuous(): controls numeric y-axis breaks, labels, limits, and spacing.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# The four bars exhaust this binomial distribution; their masses sum to one.
# Highlighting exactly two matches the business question rather than implying observed
 ↪  frequencies.
ggplot(binomial, aes(Purchases, Probability, fill = Purchases == 2)) +
  geom_col(width = .65) +
  geom_text(aes(label = sprintf("%.3f", Probability)), vjust = -.4) +
  scale_fill_manual(values = c("#B6D3E1", "#E4B47E"), guide = "none") +
  scale_x_continuous(breaks = 0:3) +
  scale_y_continuous(limits = c(0, .50)) +
  labs(x = "Purchases among three customers", y = "Probability mass",
       title = "A binomial model for Martin Clothing Store")
```



Original R model plot using the book's $n = 3$ and $p = 0.30$. These bars are probabilities, not observed customer counts.

Simulation checks an implication of the model

```
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# dbinom(): returns binomial point probabilities for size trials and success probability
 ↪  prob.
# set.seed() makes the simulated illustration repeatable; rbinom() generates counts under
 ↪  the model.
# mean(simulated == 2) estimates the event probability, not the real store's unknown
 ↪  purchase rate.
set.seed(307312)
simulated <- rbinom(10000, size = 3, prob = .30)
kable(data.frame(Method = c("Exact model", "10,000 simulated groups"),
                 Probability_of_two = c(dbinom(2, 3, .30), mean(simulated == 2))), digits
                  ↪  = 4)
```

Method	Probability_of_two
Exact model	0.1890
10,000 simulated groups	0.1907

The simulated proportion should be near 0.189 but need not equal it. More repetitions usually reduce simulation variability. Agreement with a simulation generated from the same assumptions **does not validate those assumptions for real customers**.

Business case: arrivals in an interval

The book considers arrivals at a **large hospital emergency room on weekday mornings**, with an average of **10 patients per 15-minute interval**. Treat the example as a count-model

exercise: assume a constant arrival rate, independent counts in disjoint intervals, and individual arrivals rather than batches. What is the probability of exactly five arrivals in 15 minutes, or exactly one arrival in three minutes? The counted event is one patient arrival; the exposure is a specified time interval. These stated inputs do not establish a current operational forecast.

يتناول الكتاب وصول المرضى إلى **قسم الطوارئ في مستشفى كبير صباح أيام العمل**، بمتوسط **عشرة مرضى خلال كل فترة مدتها خمس عشرة دقيقة**. نستخدم المثال للتدرّب على نموذج العدّ: نفترض ثبات معدل الوصول واستقلال الأعداد في الفترات غير المتداخلة، ووصول الأفراد دون دفعات جماعية. ما احتمال وصول خمسة مرضى بالضبط خلال خمس عشرة دقيقة، أو مريض واحد بالضبط خلال ثلاث دقائق؟ الحدث المعدود هو وصول مريض واحد، ومدة التعرّض هي الفترة الزمنية المحددة. لا تثبت هذه المعطيات توقعًا تشغيليًا حاليًا.

The Poisson model uses an interval mean

For a homogeneous Poisson process, a constant rate, independent increments, and negligible probability of multiple arrivals in a sufficiently short interval imply Poisson counts. Merely having a count variable or knowing its mean is insufficient.

If $X \sim \text{Poisson}(\lambda)$,

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \dots,$$

and $E(X) = \text{Var}(X) = \lambda$. Here, λ is the **mean count for the specified interval**. The book writes this mean as μ; R calls the argument `lambda`.

For 15 minutes, $\lambda = 10$. For three minutes at the same rate,

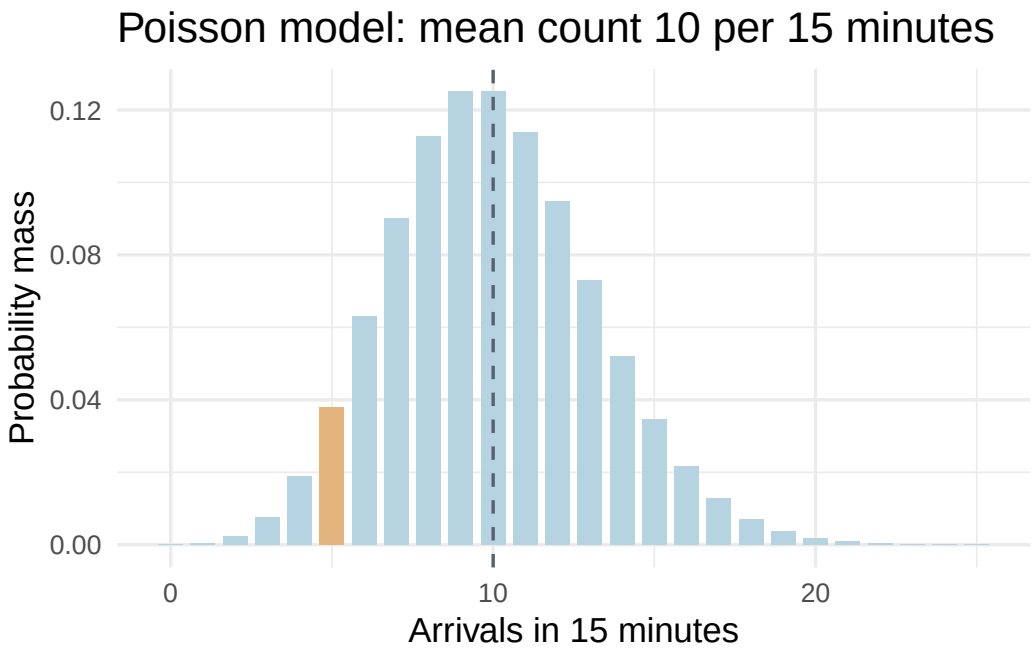
$$\lambda_{3\text{ min}} = 10 \times \frac{3}{15} = 2.$$

```
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# names(): reads or assigns names to vector elements or table columns.
# unname(): removes names or dimension names without changing the underlying values.
# dpois() returns an exact-count mass; ppois(k, lower.tail=FALSE) returns P(X>k).
# Scale lambda with the interval length before calculating a three-minute probability.
lambda_15 <- 10
lambda_3 <- lambda_15 * 3 / 15
poisson_answers <- c("Exactly 5 in 15 minutes" = dpois(5, lambda = lambda_15),
                     "More than 10 in 15 minutes" = ppois(10, lambda = lambda_15,
                      ↪  lower.tail = FALSE),
                     "Exactly 1 in 3 minutes" = dpois(1, lambda = lambda_3))
kable(data.frame(Event = names(poisson_answers), Probability = unname(poisson_answers)),
 ↪  digits = 4)
```

Event	Probability
Exactly 5 in 15 minutes	0.0378
More than 10 in 15 minutes	0.4170
Exactly 1 in 3 minutes	0.2707

The exact-five probability is about **0.0378**. The exact-one probability over three minutes is about **0.2707**. Keeping `lambda=10` for three minutes would use the wrong exposure.

```r
# data.frame(): builds a table whose named columns have compatible lengths.
# dpois(): returns Poisson count probabilities; lambda is the mean for the stated
 ↪  exposure.
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_col(): draws bars at supplied heights; it does not count records.
# geom_vline(): adds vertical references at xintercept; linetype selects solid or dashed
 ↪  lines.
# scale_fill_manual(): assigns explicit fill colors to categories; named values preserve
 ↪  category-color mappings.
# c(): combines values into a vector, converting them to a common type when needed.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# Plot a finite window of an infinite-support distribution; the omitted upper tail is not
 ↪  zero.
# The dashed line marks the model mean, while the highlighted bar marks exactly five
 ↪  arrivals.
poisson <- data.frame(Arrivals = 0:25, Probability = dpois(0:25, lambda = 10))
ggplot(poisson, aes(Arrivals, Probability, fill = Arrivals == 5)) +
  geom_col(width = .75) +
  geom_vline(xintercept = 10, linetype = "dashed", color = "#56616D") +
  scale_fill_manual(values = c("#B6D3E1", "#E4B47E"), guide = "none") +
  labs(x = "Arrivals in 15 minutes", y = "Probability mass",
       title = "Poisson model: mean count 10 per 15 minutes")
```



Original R model plot. Only counts 0–25 are displayed; the model also allows larger counts. The dashed line is the mean, not a capacity limit.

Choose by the mechanism

Question	Binomial	Poisson
What is fixed?	Number of trials	Exposure interval
What is counted?	Successes among trials	Occurrences during exposure
Parameters	n, p	Mean count λ for that exposure
Possible counts	0 through n	Any nonnegative integer

Question	Binomial	Poisson
Main assumptions	Independent trials, common p	Constant-rate process, independent increments, no batches

A small count by itself does not select a model. Discuss the process and assumptions first. Complete Lesson 4 practice, then continue to continuous distributions.

Sources and limits

Anderson et al. (2020), 8th ed., §5.5, especially p.244, and §5.6, pp.252–255. Parameters were checked directly against the local PDF. No customer records or arrival timestamps are available here. Simulation generates explicitly artificial groups under the binomial model; it does not replace missing publisher data or establish model fit.